

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 08: JOURNEY INSIDE THE ATOM: STRUCTURE & COMPOSITION
TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-08

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

- The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
- Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
- All questions are compulsory. There is no negative marking.
- Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 In Rutherford's alpha scattering experiment, what is the relationship between the number of scattered particles N and scattering angle θ ?

- (A) $N \propto \sin^4(\theta/2)$ (B) $N \propto \sin(\theta/2)$
(C) $N \propto \cos^2\theta$ (D) $N \propto \theta^2$

Q.2 What is the distance of closest approach (d) of an alpha particle of kinetic energy E_k head-on to a gold nucleus ($Z = 79$)?

- (A) $d = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{E_k}$ (B) $d = \frac{Ze^2}{E_k^2}$
(C) $d = \frac{E_k}{Ze}$ (D) $d = 2ZeE_k$

Q.3 According to Bohr's postulate, what is the quantized orbital angular momentum (L) of an electron in the n -th shell?

- (A) $L = \frac{n}{2}\pi h$ (B) $L = n\pi h$
(C) $L = \frac{h}{2\pi n}$ (D) $L = n^2 h$

Q.4 What is the wavelength (λ) of radiation emitted when an electron jumps from higher energy E_2 to lower energy E_1 (h is Planck's constant, c is speed of light)?

- (A) $\lambda = \frac{hc}{E_2 - E_1}$ (B) $\lambda = \frac{E_2 - E_1}{hc}$
(C) $\lambda = h(E_2 - E_1)$ (D) $\lambda = \frac{c}{E_2 - E_1}$

Q.5 How many neutrons are present in the nucleus of a neutral atom of Thorium-232 (${}^{232}_{90}\text{Th}$)?

- (A) 90 (B) 142
(C) 232 (D) 322

Q.6 An atom has 15 protons, 16 neutrons, and 18 electrons. What is the identity and charge of this species?

- (A) Phosphide anion (P^{3-}) (B) Phosphorus neutral atom (P)
(C) Sulfur cation (S^{2+}) (D) Argon atom (Ar)

Q.7 Which pair represents isobars?

- (A) ${}^{40}_{18}\text{Ar}$ and ${}^{40}_{20}\text{Ca}$ (B) ${}^{12}_6\text{C}$ and ${}^{14}_6\text{C}$
 (C) ${}^1_1\text{H}$ and ${}^2_1\text{H}$ (D) ${}^{35}_{17}\text{Cl}$ and ${}^{37}_{17}\text{Cl}$

Q.8 Which pair represents isotones (atoms having identical numbers of neutrons)?

- (A) ${}^{14}_6\text{C}$ and ${}^{16}_8\text{O}$ (B) ${}^{12}_6\text{C}$ and ${}^{14}_7\text{N}$
 (C) ${}^{40}_{18}\text{Ar}$ and ${}^{40}_{20}\text{Ca}$ (D) ${}^1_1\text{H}$ and ${}^2_1\text{H}$

Q.9 Naturally occurring boron consists of two isotopes: ${}^{10}\text{B}$ (mass 10 u) and ${}^{11}\text{B}$ (mass 11 u). If average atomic mass is 10.8 u, what is the abundance of ${}^{10}\text{B}$?

- (A) 20% (B) 80%
 (C) 50% (D) 10%

Q.10 What is the maximum number of electrons in an atom that can have principal quantum number $n = 4$?

- (A) 16 (B) 32
 (C) 18 (D) 64

Q.11 Which isotope is used in carbon dating to estimate the archaeological age of ancient wooden fossils?

- (A) ${}^{14}\text{C}$ (B) ${}^{12}\text{C}$
 (C) ${}^{13}\text{C}$ (D) ${}^{16}\text{O}$

Q.12 Why does a hydrogen atom ($Z=1$) exhibit multiple distinct spectral emission lines in its absorption/emission spectrum?

- (A) A hydrogen atom contains multiple electrons (B) A sample contains billions of hydrogen atoms where single electrons transition between multiple quantized energy levels ($n=1, 2, 3, 4, \dots$)
 (C) Hydrogen splits into protons (D) Hydrogen has isotopes only

Q.13 What is the charge and mass of an electron in SI units?

- (A) $-1.602 \times 10^{-19} \text{ C}$, $9.109 \times 10^{-31} \text{ kg}$ (B) $+1.602 \times 10^{-19} \text{ C}$, $1.672 \times 10^{-27} \text{ kg}$
 (C) Zero, $9.109 \times 10^{-31} \text{ kg}$ (D) $-1.602 \times 10^{-19} \text{ C}$, $1.675 \times 10^{-27} \text{ kg}$

Q.14 What is the valency of Silicon ($Z = 14$, config $2, 8, 4$)?

- (A) 4 (B) 2
 (C) 6 (D) 8

Q.15 What is the chemical formula of Aluminum Sulfate (Aluminum valency 3, Sulfate radical SO_4^{2-} valency 2)?

- (A) AlSO_4 (B) $\text{Al}_2(\text{SO}_4)_3$
 (C) $\text{Al}_3(\text{SO}_4)_2$ (D) Al_2SO_4

Q.16 What is the chemical formula of Ammonium Carbonate (Ammonium NH_4^+ , Carbonate CO_3^{2-})?

- (A) $(\text{NH}_4)_2\text{CO}_3$ (B) $\text{NH}_4(\text{CO}_3)_2$
 (C) NH_4CO_3 (D) $\text{N}_2\text{H}_8\text{CO}_3$

Q.17 Which subatomic particle has approximately zero rest mass and zero charge, and was postulated by Wolfgang Pauli to conserve energy in beta decay?

- (A) Neutrino (ν) (B) Neutron
 (C) Positron (D) Proton

Q.18 The radius of an atomic nucleus of mass number A is proportional to:

- (A) $A^{1/3}$ (B) A^3
(C) A (D) $A^{1/2}$

Q.19 What is the mass defect in a nuclear reaction?

- (A) Difference between total mass of constituent nucleons and actual bound nuclear mass ($\Delta m = \Delta m c^2$) (B) Instrumental scale error
(C) Mass of lost electrons (D) Weight of neutrons

Q.20 Which element has atomic number 19 and config $2, 8, 8, 1$?

- (A) Potassium (K) (B) Sodium (Na)
(C) Calcium (Ca) (D) Argon (Ar)

Q.21 Why is the average atomic mass of a chemical element rarely an exact whole number integer?

- (A) Because mass of an electron fluctuates (B) Because natural elements exist as mixtures of isotopes with different mass numbers in varying percentage abundances
(C) Because measuring balances are inaccurate (D) Because neutrons dissolve

Q.22 Which isotope of hydrogen is radioactive and decays by beta emission?

- (A) Tritium (3_1H) (B) Protium (1_1H)
(C) Deuterium (2_1H) (D) All isotopes

Q.23 What is heavy water chemically?

- (A) Deuterium oxide (D_2O or 2H_2O) (B) Water saturated with salt
(C) Water under 100 atmospheres (D) Hydrogen peroxide (H_2O_2)

Q.24 What did J.J. Thomson's 'plum pudding' model fail to explain?

- (A) Large angle deflections of alpha particles in Rutherford's scattering experiment (B) Electrical neutrality of atoms
(C) Presence of electrons (D) Mass of atoms

Q.25 Which quantum energy state corresponds to the lowest possible energy configuration of an electron in an atom?

- (A) Ground state ($n=1$) (B) Excited state
(C) Ionized state (D) Stationary transition

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(A)	Rutherford's classical scattering differential cross-section formula: $N(\theta) \propto \csc^4(\theta/2) = \frac{1}{\sin^4(\theta/2)}$.
Q.2	(A)	At closest approach, kinetic energy converts to electrostatic potential energy: $E_k = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{d} \Rightarrow d = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{E_k}$.
Q.3	(A)	Bohr's quantum condition: Angular momentum is an integral multiple of $\hbar/2\pi$ ($mvr = \frac{nh}{2\pi}$).
Q.4	(A)	$\Delta E = E_2 - E_1 = h\nu = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E_2 - E_1}$.
Q.5	(B)	Number of neutrons $N = A - Z = 232 - 90 = 142$.
Q.6	(A)	$Z = 15$ identifies Phosphorus. It has 3 extra electrons ($18 - 15 = 3$), giving a trivalent anion P^{3-} .
Q.7	(A)	Isobars have different atomic numbers but identical mass numbers ($A = 40$ in Argon and Calcium).
Q.8	(A)	In ${}^{14}_6\text{C}$, $N = 14 - 6 = 8$. In ${}^{16}_8\text{O}$, $N = 16 - 8 = 8$. Identical neutron count defines isotones.
Q.9	(A)	$10x + 11(1 - x) = 10.8 \Rightarrow 10x + 11 - 11x = 10.8 \Rightarrow 11 - x = 10.8 \Rightarrow x = 0.20 = 20\%$.
Q.10	(B)	Capacity $= 2n^2 = 2(4)^2 = 2 \times 16 = 32$ electrons.
Q.11	(A)	Carbon-14 (${}^{14}_6\text{C}$) is a beta-emitting radioactive isotope with a half-life of 5,730 years used in radiocarbon dating.
Q.12	(B)	Different atoms in the macroscopic gas sample undergo electronic transitions across different energy levels, producing distinct spectral lines.
Q.13	(A)	An electron has fundamental negative charge $e = -1.602 \times 10^{-19}$ C and rest mass 9.109×10^{-31} kg.
Q.14	(A)	Silicon shares 4 valence electrons to complete its octet, exhibiting a valency of 4 (tetravalent).
Q.15	(B)	Criss-crossing valencies (Al^{3+} and SO_4^{2-}) gives $\text{Al}_2(\text{SO}_4)_3$.
Q.16	(A)	Criss-crossing valencies (NH_4^{1+} and CO_3^{2-}) gives $(\text{NH}_4)_2\text{CO}_3$.
Q.17	(A)	Neutrinos are neutral leptons with near-zero rest mass emitted during radioactive beta decay.
Q.18	(A)	Nuclear radius $R = R_0 A^{1/3}$, where $R_0 \approx 1.2 \times 10^{-15}$ m (1.2 fermi).
Q.19	(A)	Mass defect Δm represents binding energy released when protons and neutrons bind to form a nucleus.
Q.20	(A)	Potassium (K , $Z = 19$) has 19 electrons arranged as 2, 8, 8, 1.
Q.21	(B)	Weighted averaging across multiple isotopic fractions yields fractional atomic weights (e.g., $\text{Cl} = 35.5$ u).
Q.22	(A)	Tritium (${}^3_1\text{H}$) has 1 proton and 2 neutrons ($N/Z = 2$), making it an unstable beta-emitter with half-life 12.3 years.
Q.23	(A)	Heavy water is deuterium oxide (D_2O), where hydrogen is replaced by its heavier isotope deuterium (${}^2_1\text{H}$).
Q.24	(A)	Thomson's diffuse positive cloud could not exert the concentrated Coulomb repulsion required to deflect alpha particles at large angles.
Q.25	(A)	The ground state ($n=1$, K-shell) is the lowest energy, thermodynamically most stable electronic state.